NAME : .....

ALLEN CAREER INSTITUTE KOTA (RAJASTHAN)
RACE Regular Analysis through Continuous Exercise
JEE (Main + Advanced) 2024
ENTHUSIAST + LEADER COURSE

SCORE ADVANCED	MATHEMATICS	M.M. 85
REVISION ASSINGMENT # 07	(CONIC SECTION-01)	TIME : 75 MIN.

SECTION–I

Straight Objective Type (3 Marks each, –1 for wrong answer)

1. Let $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ and $\frac{x^2}{A^2}-\frac{y^2}{B^2}=1$ are confocal $(a>A \ \& \ a>b)$ having the foci at S_1 & S_2. Let P be their point of intersection, then S_1P & S_2P are the roots of quadratic equation -

(A) $x^2 + 2ax + (a^2 – A^2) = 0$ (B) $x^2 – 2Ax + (a^2 + A^2) = 0$

(C) $x^2 – 2Ax + (a^2 – A^2) = 0$ (D) $x^2 – 2ax + (a^2 – A^2) = 0$

2. L_1 is a tangent drawn to the curve $x^2 – 4y^2 = 16$ at $A\left(5, \frac{3}{2}\right)$. L_2 is another tangent parallel to L_1 which meets the curve at B. L_3 and L_4 are normals to the curve at A and B. Lines L_1, L_2, L_3, L_4 forms a rectangle, then which of the following is correct

(A) Equation of tangent at B is $6y = 5x – 16$.

(B) Equation of normal at B is $12x + 10y + 74 = 0$

(C) Radius of largest circle inscribed in the rectangle is $\frac{32}{\sqrt{61}}$.

(D) Radius of the circle circumscribing the rectangle is $\frac{\sqrt{109}}{2}$

3. From a point on the line $x – y + 2 = 0$ tangents are drawn to the hyperbola $\frac{x^2}{6}-\frac{y^2}{2}=1$, such that the chord of contact passes through a fixed point (λ, μ). Then $\frac{\lambda}{\mu}$ is equal to -

(A) 2 (B) 3 (C) 4 (D) 5

4. The points of contact Q and R of tangents from the point P(2, 3) to the parabola $y^2 = 4x$ are -

(A) (9, 6) and (1, 2) (B) (4, 4) and (9, 6) (C) (9, 6) and $\left(\frac{1}{4}, 1\right)$ (D) (1, 2) and (4, 4)

Space for Rough Work

For More Material Join: @JEEAdvanced_2025

5. If $(3m_i^2, -6m_i)$ represents the feet of the normals to the parabola $y^2 = 12x$ from (1, 2), then $\sum\frac{1}{m_i}$ is -

(A) $-\frac{5}{2}$ (B) $\frac{3}{2}$ (C) 6 (D) – 3

6. The common tangent of two parabolas $y = x^2$ and $y = x^2 - 2x + 2$ is -

(A) $y = 2x - 1$ (B) $4y - x + 4 = 0$ (C) $4x - 4y - 1 = 0$ (D) $y + 4x + 4 = 0$

7. The locus of the point of intersection of perpendicular tangents to the parabola $x^2 - 8x + 2y + 2 = 0$ is -

(A) $2y - 15 = 0$ (B) $2y + 15 = 0$ (C) $2x + 9 = 0$ (D) none of these

8. The equation to the locus of the mid point of the line segment joining the focus of the parabola $y^2 = 8x$ to a moving point on the given parabola is -

(A) $y^2 = 4x - 1$ (B) $y^2 = 4x - 4$ (C) $x^2 = 4y - 1$ (D) $x^2 + y^2 = 1$

9. If lines $3x + 5y = 20$ and $3x - 5y = 20$ are tangents at the extremities of a latus rectum to an ellipse whose centre is origin, then length of the latus rectum is -

(A) 0.16 (B) 2.56 (C) 3.2 (D) 5.12

10. If the ellipse $\frac{x^2}{16} + \frac{y^2}{b^2} = 1$ is inscribed in a square of side length $5\sqrt{2}$, then area of ellipse is -

(A) 4π (B) 8π (C) 10π (D) 12π

Multiple Correct Answer Type (4 Marks each, –1 for wrong answer)

11. Equation of common tangent(s) to $x^2 - y^2 = 12$ and $xy = 8$ is (are) -

(A) $y = 3x + 4\sqrt{6}$ (B) $y = -3x + 4\sqrt{6}$ (C) $3y = x + 4\sqrt{6}$ (D) $y = -3x - 4\sqrt{6}$

12. $y = f(x)$ is a parabola having directrix with slope 0.

If $f^2(0) + f^2(1) + f^2(2) + 45 = 10f(0) + 8f(1) + 4f(2)$, then -

(A) $y = f(x)$ is symmetric about $x = -\frac{1}{2}$ (B) $f(x) < 0 \ \forall \ x \in R$

(C) $\ell n|(f(x))|$ is one-one function (D) range of $f(x)$ is $\left(-\infty, \frac{41}{8}\right]$

13. For the parabola $x^2 + y^2 + 2xy - 4x + 4y - 4 = 0$ -

(A) The focus is (0,0) (B) The focus is $\left(-\frac{1}{2}, \frac{1}{2}\right)$

(C) The directrix is $x - y + 2 = 0$ (D) The directrix is $x - y - 2 = 0$

Space for Rough Work

For More Material Join: @JEEAdvanced_2025

14. If one of the foci of an ellipse is (2, 3) whose eccentricity is $\frac{1}{3}$ and point of intersection of its major axis with the corresponding directrix is (10, 11), then identify the correct statement(s) -

(A) vertices of the ellipse are (–2, –1) and (4, 5) (B) major axis is $6\sqrt{2}$

(C) minor axis is 8 (D) latus rectum is $\frac{32\sqrt{2}}{3}$

15. Let an incident ray $L_1 = 0$ gets reflected at point A(–2, 3) on hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ & passes through focus S(2, 0), then -

(A) equation of incident ray is $x + 2 = 0$ (B) equation of reflected ray is $3x + 4y = 6$

(C) eccentricity, $e = 2$ (D) length of latus rectum = 6

16. Consider four points $\left(\pm\frac{ab}{\sqrt{a^2+b^2}}, \pm\frac{ab}{\sqrt{a^2+b^2}}\right)$ which of the following is correct :-

(A) Only two ellipse can pass through them (B) Only one ellipse can pass through them

(C) No parabola can pass through them (D) Exactly two parabolas will pass through them

Linked Comprehension Type (Single Correct Answer Type) (3 Marks each, –1 for wrong answer)

Paragraph for Question 17 & 18

Let minimum value of $f(x) = |x - a| + 2|x - 2|$ is 1 $a > 2$. Consider an ellipse $\frac{x^2}{a^2} + \frac{y^2}{4} = 1$. Tangents are drawn from point P(1, 2) to the ellipse.

On the basis of above information answer the following :

17. If 'e' is eccentricity of given ellipse, then e can be -

(A) $\frac{\sqrt{3}}{2}$ (B) $\frac{\sqrt{5}}{3}$ (C) $\frac{1}{3}$ (D) None of these

18. The point of tangency of tangents from P is point Q where eccentric angle is θ, then

(A) $\sin\theta = 1$ (B) $\sec\theta = \frac{5}{3}$ (C) $\tan\theta = \frac{4}{3}$ (D) None of these

Space for Rough Work

For More Material Join: @JEEAdvanced_2025

Paragraph for Question 19 & 20

Consider three curves

$C_1 : x^2 - y^2 = 1 \ (x < 0)$

$C_2 : xy = 1 \ (y > 0)$

C_3 : Ellipse whose one focus is also focus of C_2 and corresponding directrix of this focus is asymptote of curve C_1.

Also length of major axis of C_3 is $\frac{8}{3}$.

On the basis of above information, answer the following questions :

19. Eccentricity of C_3 is equal to -

(A) $\frac{1}{4}$ (B) $\frac{1}{2\sqrt{2}}$ (C) $\frac{1}{\sqrt{2}}$ (D) $\frac{1}{2}$

20. If $A\left(-\frac{1}{2}, -2\right)$ and BC is double ordinate of curve C_2 then locus of orthocentre of ΔABC is a -

(A) fixed point (B) parabola (C) ellipse (D) hyperbola

SECTION–III

Numerical Grid Type (Single digit Ranging from 0 to 9) (4 Marks each, –1 for wrong answer)

1. If the image of the parabola $y = x^2$ in the line $x + y = 1$ is $x + y^2 = ky$, then k is equal to

2. From a variable point P, two tangents are drawn to the parabola $y^2 = 4ax$ and these tangents meet the co-ordinate axes in concyclic points. If locus of P is $x = \lambda_1 a$, then λ_1 is equal to

3. Let PQ is a tangent to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ meeting the positive x & y axis at points P & Q respectively.

Point R divides PQ internally in the ratio 2 : 1. If locus of R is $\frac{a^2}{x^2} + \frac{4b^2}{y^2} = \lambda$, then λ is equal to

4. The slope of one of the common tangents to the hyperbolas $\frac{x^2}{16} - \frac{y^2}{9} = 1$ and $\frac{y^2}{16} - \frac{x^2}{9} = 1$ is m, then |m| is

5. The chord of the hyperbola $x^2 - y^2 = 25$, touches the parabola $y^2 = 20x$. If the locus of middle point of the chords can be expressed as $x^3 = y^2 (x - a)$, then value of a is

Space for Rough Work

	Q.	1	2	3	4	5	6	7	8	9	10
SECTION-I	A.	C	B	A	B	C	B	D	D	A	B
	Q.	11	12	13	14	15	16	17	18		
	A.	C,D	A,B,D	B,C,D	B,D	B	C	C	B		
SECTION-III	Q.	1	2	3	4						
	A.	2	7	9	5						
SECTION-IV	Q.1	A	B	C	D						
		S	R	P	Q						

For More Material Join: @JEEAdvanced_2025